SLEEVEDRIFT · FEASIBILITY ASSESSMENT

SleeveDrift feasibility assessment

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DOCUMENT CONTROL

Document ID	SVD-FEA-001	Version	0.1
Status	Draft	Date	2026-11-02
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/sleevedrift		
Source	docs/08-feasibility.md @ 1ba9c34		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-02	Amish Chadha	First issue at the kit 1.9.1 upgrade, for option A

SleeveDrift feasibility assessment

Option A is feasible on paper and buildable by a general steel workshop; what stays uncertain is whether it holds up in a real collapse pipe (grit, rope wear, ring friction), which only trials at TRL 4 can show, and it costs about twice the value-engineering target. Method: STANDARDS section 23.

Table 1. Feasibility by aspect.

Aspect	Assessment	Rating	Risks and notes	Evidence
Design	Ten STEM claims are within known science, eleven are plausible and one (S17, the casualty rope factor of 6.5 against 15) fails with a fix proposed to Amish; every requirement is met on paper except cost (PR-013), with PR-009 managed by inspection	High	The margins that are thin: the haul rate (0.42 against 0.4 m3/h), the heaviest package (37.7 against 40 kg), the set-up time (1.6 against 2 h)	SVD-STM-001, SVD-CAL-001
Materials	S355 plate and section, galvanised sheet, UHMW-PE runners, polyester rope, layflat duct; all stock materials	High	Pipe wall and weld beads at a real site may differ from the 780 mm bore assumed	SVD-CAL-001 A, BOM
Manufacturing and assembly	Rolled shell segments, flat-bar flanges, bent sheet sledges and welded SHS trestles; 153 made and bought parts in the detailed model; four segments go in one at a time on a sledge	Medium	Rolling 5 mm plate to a 760 mm ring and keeping the four segments to a 20 mm joint tolerance needs a workshop with plate rolls	SVD-BLD-001, cad/fidelity.json
Power and controls	Nothing powered in the pipe; battery drills and a 230 V blower at the portal; two sound-powered telephones	High	Portal power may be a generator; clutches are set at the portal	PR-008, SVD-CAL-001 D, F
Supply of parts	Winches, drills, sheave, monitors, blower and duct are catalogue items; the one long-lead item is the bought roll-up stretcher	Medium	Prices are first-pass estimates from Amish's earlier sessions; two lines (the signal line at USD 860, the winches at USD 2,300) are the least tested	BOM
Cost	USD 10,804.00 of parts against the USD 5,000 target (PR-013)	Low	A kit for rescue is not a consumer product; the target is a control figure (SVD-DDR-001 item 13). Proposed, awaiting Amish: hold it or restate it	BOM, SVD-CAL-001 J
Safety case	Overload limit, hood, gas monitoring and guards are designed in	Medium	Hazards that remain: face collapse, rope whip, drill torque on the winch, entrapment (SVD-RSK-001); none is closed by tests yet	SVD-RSK-001, SVD-CAL-001
Regulatory	Published as a reference, so none to meet; operators meet confined-space rules	Medium	Rope marking (NFPA 1983 practice) and flammable-atmosphere use to assess	SVD-REG-001
Concept options	A scored 175 of 195, B 164 and C 148; A and B are within 10 %	Medium	Both go to Amish; the choice is open	SVD-OPT-001

Ratings: High, Medium, Low.